
LAB SESSION 05

PROGRAM FLOW CONTROL USING JUMP

1. Write a program to display the first letter of your name 10 times using jump instruction. For example, your name starts with "A" then the output will be:

AAAAAA

Source Code:

```
01 TITLE LAB 5 ACTIVITY 1      ;Title of the program
02 .MODEL SMALL                ;Define memory model
03 .STACK 100H                 ;Set the size of the stack
04 .CODE                       ;Code segment Starts
05
06 MAIN PROC
07     MOV CX, 10                ;Initialize CL With The Count (10)
08     MOV AL, 'S'              ;Initialize AL With The Character 'S'
09 LOOP1:
10     MOV AH, 2                 ;Set AH to 2, indicating the "print character" function
11     MOV DL, AL                ;Move the character from AL to DL
12     INT 21H                   ;Call the DOS interrupt to print the character
13
14     DEC CX                    ;Decrement the loop counter
15
16     JNZ LOOP1                 ;Jump back to LOOP1 if CX is not zero
17
18     MOV AH, 4CH               ;Set AH to 4CH, indicating the "terminate program" function
19     INT 21H                   ;Call the DOS interrupt to exit to DOS
20
21 MAIN ENDP                    ;End Of The Main Procedure
22 END MAIN                     ;End Of The Entire Program
```

OUTPUT:

 emulator screen (80x25 chars)



2. Write a program to display all the ASCII characters on screen line by line. Make sure to display character screen-by-screen.

(Hint: After displaying first 23 characters, ask user to continue by providing the statement "press any key to continue..." to display next 23 characters and so on.)

Source Code:

```

001 TITLE LAB 5 ACTIVITY 2      ;Title of the program
002 .MODEL SMALL                ;Define memory model
003 .STACK 100H                 ;Set the size of the stack
004 .DATA                       ;Declare a section of memory for storing data
005
006 PROMPT DB 'THE 256 ASCII CHARACTER ARE: $'
007 CONTINUE DB 'PRESS ANY KEY TO CONTINUE... $'
008
009 .CODE                        ;Code Segment starts
010 MAIN PROC
011
012     ; INITIATE DATA SEGMENT
013     MOV AX, @DATA
014     MOV DS, AX
015
016     ; DISPLAY PROMPT
017     MOV AH, 9
018     MOV DX, OFFSET PROMPT
019     INT 21H
020
021     ; NEXT LINE
022     MOV AH, 2
023     MOV DL, 0AH
024     INT 21H
025
026     ; MOVE THE CURSOR TO THE BEGINNING OF THE CURRENT LINE
027     MOV DL, 0DH
028     INT 21H
029
030     MOV CX, 23
031     ; Initialize DL to 0
032     MOV DL, 0
033
034 LOOP1:
035     MOV AH, 2
036     INT 21H
037     ;Increment DL
038     INC DL
039     ;Decrement CX
040     DEC CX
041     JNZ LOOP1
042
043     MOV AH, 2
044     MOV DL, 0AH
045     INT 21H
046
047     MOV DL, 0DH
048     INT 21H
049
050     ; Display continue message
051     MOV AH, 9
052     MOV DX, OFFSET CONTINUE
053     INT 21H
054
055     ; Wait for user input
056     MOV AH, 1
057     INT 21H

```

```
058
059 ; Clear screen
060 MOV AH, 6
061 MOV AL, 0
062 MOV BH, 7
063 MOV CX, 0
064 MOV DX, 184FH
065 INT 10H
066
067 MOV CX, 23
068 MOV DL, 18H
069
070 LOOP2:
071     MOV AH, 2
072     INT 21H
073
074     INC DL
075
076     DEC CX
077     JNZ LOOP2
078
079     MOV AH, 2
080     MOV DL, 0AH
081     INT 21H
082
083     MOV DL, 0DH
084     INT 21H
085
086 ; Display continue message
087 MOV AH, 9
088 MOV DX, OFFSET CONTINUE
089 INT 21H
090
091 ; Wait for user input
092 MOV AH, 1
093 INT 21H
094
095 ; Clear screen
096 MOV AH, 6
097 MOV AL, 0
098 MOV BH, 7
099 MOV CX, 0
100 MOV DX, 184FH
101 INT 10H
102
103 MOV CX, 23
104 MOV DL, 2FH
105
106 LOOP3:
107     MOV AH, 2
108     INT 21H
109     INC DL
110     DEC CX
111     JNZ LOOP3
112
```

```
113     MOV AH, 2
114     MOV DL, 0AH
115     INT 21H
116     MOV DL, 0DH
117     INT 21H
118
119     ; DISPLAY CONTINUE MESSAGE
120     MOV AH, 9
121     MOV DX, OFFSET CONTINUE
122     INT 21H
123
124     ; WAIT FOR USER INPUT
125     MOV AH, 1
126     INT 21H
127
128     ; CLEAR SCREEN
129     MOV AH, 6
130     MOV AL, 0
131     MOV BH, 7
132     MOV CX, 0
133     MOV DX, 184FH
134     INT 10H
135
136     MOV CX, 23
137     MOV DL, 46H
138
139 LOOP4:
140     MOV AH, 2
141     INT 21H
142     INC DL
143     DEC CX
144     JNZ LOOP4
145
146     MOV AH, 2
147     MOV DL, 0AH
148     INT 21H
149     MOV DL, 0DH
150     INT 21H
151
152     ; DISPLAY CONTINUE MESSAGE
153     MOV AH, 9
154     MOV DX, OFFSET CONTINUE
155     INT 21H
156
157     ; WAIT FOR USER INPUT
158     MOV AH, 1
159     INT 21H
160
161     ; CLEAR SCREEN
162     MOV AH, 6
163     MOV AL, 0
164     MOV BH, 7
165     MOV CX, 0
166     MOV DX, 184FH
167     INT 10H
```

```
168
169     MOV CX, 23
170     MOV DL, 5DH
171
172 LOOP5:
173     MOV AH, 2
174     INT 21H
175     INC DL
176     DEC CX
177     JNZ LOOP5
178
179     MOV AH, 2
180     MOV DL, 0AH
181     INT 21H
182     MOV DL, 0DH
183     INT 21H
184
185     ; DISPLAY CONTINUE MESSAGE
186     MOV AH, 9
187     MOV DX, OFFSET CONTINUE
188     INT 21H
189
190     ; WAIT FOR USER INPUT
191     MOV AH, 1
192     INT 21H
193
194     ; CLEAR SCREEN
195     MOV AH, 6
196     MOV AL, 0
197     MOV BH, 7
198     MOV CX, 0
199     MOV DX, 184FH
200     INT 10H
201
202     MOV CX, 23
203     MOV DL, 74H
204
205 LOOP6:
206     MOV AH, 2
207     INT 21H
208     INC DL
209     DEC CX
210     JNZ LOOP6
211
212     MOV AH, 2
213     MOV DL, 0AH
214     INT 21H
215     MOV DL, 0DH
216     INT 21H
217
218     ; DISPLAY CONTINUE MESSAGE
219     MOV AH, 9
220     MOV DX, OFFSET CONTINUE
221     INT 21H
222
223     ; WAIT FOR USER INPUT
224     MOV AH, 1
```

```
225     INT 21H
226
227     ; CLEAR SCREEN
228     MOV AH, 6
229     MOV AL, 0
230     MOV BH, 7
231     MOV CX, 0
232     MOV DX, 184FH
233     INT 10H
234
235     MOV CX, 23
236     MOV DL, 8BH
237
238     LOOP7:
239     MOV AH, 2
240     INT 21H
241     INC DL
242     DEC CX
243     JNZ LOOP7
244
245     MOV AH, 2
246     MOV DL, 0AH
247     INT 21H
248     MOV DL, 0DH
249     INT 21H
250
251     ; DISPLAY CONTINUE MESSAGE
252     MOV AH, 9
253     MOV DX, OFFSET CONTINUE
254     INT 21H
255
256     ; WAIT FOR USER INPUT
257     MOV AH, 1
258     INT 21H
259
260     ; CLEAR SCREEN
261     MOV AH, 6
262     MOV AL, 0
263     MOV BH, 7
264     MOV CX, 0
265     MOV DX, 184FH
266     INT 10H
267
268     MOV CX, 23
269     MOV DL, 0A2H
270
271     LOOP8:
272     MOV AH, 2
273     INT 21H
274     INC DL
275     DEC CX
276     JNZ LOOP8
277
278     MOV AH, 2
279     MOV DL, 0AH
280     INT 21H
281     MOV DL, 0DH
```

```
282     INT 21H
283
284     ; DISPLAY CONTINUE MESSAGE
285     MOV AH, 9
286     MOV DX, OFFSET CONTINUE
287     INT 21H
288
289     ; WAIT FOR USER INPUT
290     MOV AH, 1
291     INT 21H
292
293     ; CLEAR SCREEN
294     MOV AH, 6
295     MOV AL, 0
296     MOV BH, 7
297     MOV CX, 0
298     MOV DX, 184FH
299     INT 10H
300
301     MOV CX, 23
302     MOV DL, 0B8H
303
304     LOOP9:
305     MOV AH, 2
306     INT 21H
307     INC DL
308     DEC CX
309     JNZ LOOP9
310
311     MOV AH, 2
312     MOV DL, 0AH
313     INT 21H
314     MOV DL, 0DH
315     INT 21H
316
317     ; DISPLAY CONTINUE MESSAGE
318     MOV AH, 9
319     MOV DX, OFFSET CONTINUE
320     INT 21H
321
322     ; WAIT FOR USER INPUT
323     MOV AH, 1
324     INT 21H
325
326     ; CLEAR SCREEN
327     MOV AH, 6
328     MOV AL, 0
329     MOV BH, 7
330     MOV CX, 0
331     MOV DX, 184FH
332     INT 10H
333
334     MOV CX, 23
335     MOV DL, 0B9H
336
```

```
337 LOOP10:
338     MOV AH, 2
339     INT 21H
340     INC DL
341     DEC CX
342     JNZ LOOP10
343
344     MOV AH, 2
345     MOV DL, 0AH
346     INT 21H
347     MOV DL, 0DH
348     INT 21H
349
350     ; DISPLAY CONTINUE MESSAGE
351     MOV AH, 9
352     MOV DX, OFFSET CONTINUE
353     INT 21H
354
355     ; WAIT FOR USER INPUT
356     MOV AH, 1
357     INT 21H
358
359     ; CLEAR SCREEN
360     MOV AH, 6
361     MOV AL, 0
362     MOV BH, 7
363     MOV CX, 0
364     MOV DX, 184FH
365     INT 10H
366
367     MOV CX, 23
368     MOV DL, 0D0H
369
370 LOOP11:
371     MOV AH, 2
372     INT 21H
373     INC DL
374     DEC CX
375     JNZ LOOP11
376
377     MOV AH, 2
378     MOV DL, 0AH
379     INT 21H
380     MOV DL, 0DH
381     INT 21H
382
383     ; DISPLAY CONTINUE MESSAGE
384     MOV AH, 9
385     MOV DX, OFFSET CONTINUE
386     INT 21H
387
388     ; WAIT FOR USER INPUT
389     MOV AH, 1
390     INT 21H
391
392     ; CLEAR SCREEN
393     MOV AH, 6
```



```

394     MOV AL, 0
395     MOV BH, 7
396     MOV CX, 0
397     MOV DX, 184FH
398     INT 10H
399
400     MOV CX, 23
401     MOV DL, 0E7H
402
403 LOOP12:
404     MOV AH, 2
405     INT 21H
406     INC DL
407     DEC CX
408     JNZ LOOP12
409
410     MOV AH, 2
411     MOV DL, 0AH
412     INT 21H
413     MOV DL, 0DH
414     INT 21H
415
416     ; DISPLAY CONTINUE MESSAGE
417     MOV AH, 9
418     MOV DX, OFFSET CONTINUE
419     INT 21H
420
421     ; WAIT FOR USER INPUT
422     MOV AH, 1
423     INT 21H
424     ; CLEAR SCREEN
425     MOV AH, 6
426     MOV AL, 0
427     MOV BH, 7
428     MOV CX, 0
429     MOV DX, 184FH
430     INT 10H
431     MOV CX, 3
432     MOV DL, 0FEH
433 LOOP13:
434     MOV AH, 2
435     INT 21H
436     INC DL
437     DEC CX
438     JNZ LOOP13
439     ;EXIT TO DOS
440     MOV AH, 4CH
441     INT 21H
442
443 MAIN ENDP                ;End Of The Main Procedure
444 END MAIN                 ;End Of The Entire Program

```

OUTPUT:

 emulator screen (80x25 chars)

```

THE 256 ASCII CHARACTER ARE:
 00000000
!@#$%^&*~
PRESS ANY KEY TO CONTINUE... _

```

↑↓→←_⌘▲ !"#%&'()*+,-.
PRESS ANY KEY TO CONTINUE... _

/0123456789:;<=>?@ABCDE
PRESS ANY KEY TO CONTINUE...

FGHIJKLMNOPQRSTUVWXYZ[\
PRESS ANY KEY TO CONTINUE...

l^ 'abcdefghijklmnopqrs
PRESS ANY KEY TO CONTINUE...

tuvwxyz{|}~ΔÇüéääâçêë
PRESS ANY KEY TO CONTINUE...

ïîïäåÉæfôöòûÿöÜ#E¥Œfáí
PRESS ANY KEY TO CONTINUE...

óúññªº¿--½¼i«»■||+|_|
PRESS ANY KEY TO CONTINUE...

⌈|⌋⌌⌍⌎⌏⌐⌑⌒⌓⌔⌕⌖⌗⌘⌙
PRESS ANY KEY TO CONTINUE... _

⌈|⌋⌌⌍⌎⌏⌐⌑⌒⌓⌔⌕⌖⌗⌘⌙
PRESS ANY KEY TO CONTINUE...

⌈|⌋⌌⌍⌎⌏⌐⌑⌒⌓⌔⌕⌖⌗⌘⌙
PRESS ANY KEY TO CONTINUE... _

τϕθΩδ∞φ€Π≡±><|÷≈°•.√²
PRESS ANY KEY TO CONTINUE...

- 3. Write a program to display the first letter of your name, using the character "*", at the center of screen. For example, if the first letter of your name is S, then following pattern should be displayed.**

Source Code:

```

001 TITLE LAB 5 ACTIVITY 3      ;Title of the program
002 .MODEL SMALL                ;Define memory model
003 .STACK 100H                 ;Set the size of the stack
004 .CODE                       ;Code segment Starts
005
006 MAIN PROC                   ;Start Main Procedure
007     MOV AX, 2                ;To Set Video Mode 2 (Monochrome)
008     INT 10H
009
010     MOV DX, 0727H            ;To Set display page and Cursor Position
011     MOV AH, 2
012     INT 10H
013
014     MOV CX, 4                ;To Initialize Loop Counter
015
016 LOOP1:
017     MOV AH, 2                ;To Display '*'
018     MOV DL, '*'
019     INT 21H
020
021     MOV AH, 2                ;Move To The Next Line
022     MOV DL, 0AH
023     INT 21H
024
025
026     MOV AH, 2                ;Move Cursor To The Beginning Of The Line
027     MOV DL, 08H
028     INT 21H
029
030     DEC CX                   ;To Decrement Loop Counter
031
032
033     JNZ LOOP1                ;Repeat Loop1 Until CX Become Zero
034
035
036     MOV DX, 0727H            ;To Set display page and Cursor Position Again
037     MOV AH, 2
038     INT 10H
039
040     MOV CX, 7                ;To Set A New Loop Counter
041
042 LOOP2:
043     MOV AH, 2                ;To Display '*'
044     MOV DL, '*'
045     INT 21H
046
047     DEC CX                   ;To Decrement Loop Counter
048
049     JNZ LOOP2                ;Repeat Loop2 Until CX Become Zero
050
051     MOV DX, 0B27H            ;To Set display page and Cursor Position Again
052     MOV AH, 2
053     INT 10H
054

```

```

054
055     MOV CX, 7           ;To Set A New Loop Counter
056
057     MOV DX, 0A2DH      ;To Set display page and Cursor Position Again
058     MOV AH, 2
059     INT 10H
060
061     MOV CX, 4           ;To Set A New Loop Counter
062
063 LOOP3:
064     MOV AH, 2           ;To Display '*'
065     MOV DL, '*'
066     INT 21H
067
068     MOV AH, 2           ;Move To The Next Line
069     MOV DL, 0AH
070     INT 21H
071
072     MOV AH, 2           ;Move Cursor To The Beginning Of The Line
073     MOV DL, 08H
074     INT 21H
075
076     DEC CX              ;To Decrement Loop Counter
077
078     JNZ LOOP3           ;Repeat Loop3 Until CX Become Zero
079
080     MOV DX, 0A27H      ;To Set display page and Cursor Position Again
081     MOV AH, 2
082     INT 10H
083
084     MOV CX, 7           ;To Set A New Loop Counter
085
086 LOOP4:
087     MOV AH, 2           ;To Display '*'
088     MOV DL, '*'
089     INT 21H
090
091     MOV AH, 2           ;Move To The Next Line
092     MOV DL, 0H
093     INT 21H
094
095     MOV AH, 2           ;Move Cursor To The Beginning Of The Line
096     MOV DL, 08H
097     INT 21H
098
099     DEC CX              ;To Decrement Loop Counter
100
101     JNZ LOOP4           ;Repeat Loop4 Until CX Become Zero
102
103
104     MOV DX, 0D27H      ;To Set display page and Cursor Position Again
105     MOV AH, 2
106     INT 10H
107
108     MOV CX, 7           ;To Set A New Loop Counter
109
110 LOOP5:
111     MOV AH, 2           ;To Display '*'
112     MOV DL, '*'
113     INT 21H
114
115     DEC CX              ;To Decrement Loop Counter
116
117     JNZ LOOP5           ;Repeat Loop4 Until CX Become Zero
118
119     MOV AH, 4CH         ;Exit To DOSBOX
120     INT 21H
121
122 MAIN ENDP              ;End Of The Main Procedure
123 END MAIN               ;End Of The Entire Program

```

OUTPUT:



4. Fill out the missing value, using target address calculation, in the following table.

IP	Opcode	Code
E06C	733F	JNZ CODE1
E06E		MOV AX,3456H
.....		
E072	7139	JNO CODE1
E074		MOV CX, 100H
.....		
E08C	8ED8	C8: MOV DS, AX
.....		
EOA7	EBE3	JMP C8
E0A9		CMP AX
.....		
E0AD	F4	CODE1: HLT